Singapore rainfall: temporal-spatial patterns and prediction

Tasks 1 and 2: Spark exploration and probability prediction, 2017–2024

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Study objectives. Using Singapore rainfall observations from 2017–2024, we analyse temporal and spatial patterns and develop models to predict the probability that station rainfall accumulated over the next 30 minutes exceeds a given threshold.

Data and methods. The eight NEA / data.gov.sg annual datasets contain 48,538,713 source records and 48,538,627 unique valid observations from 91 station IDs. Spark performs full-data cleaning and aggregation. Annual comparisons use 21 stable stations to reduce changes caused by the monitoring network.

Main findings. Of all valid five-minute observations, 97.08% are dry. Eligible diurnal samples show the highest rain frequency at 16:00–17:00 Singapore time. Available-year monthly averages have the highest frequency in November and the lowest in July, with varying eligible years and stations. In the fixed panel, duration-normalized rainfall is lowest in 2019 and highest in 2021. Nearby stations generally have more strongly correlated daily rainfall; eight years alone do not establish long-term climate change.

Year	Rain / observed 24h (mm)	Wet frequency (%)	Wet-interval intensity (mm/5min)	Mean coverage (%)
2017	6.85	2.85	0.833	94.6
2018	7.10	2.90	0.840	82.9
2019	4.81	1.97	0.841	94.7
2020	7.40	2.92	0.877	95.3
2021	7.84	3.04	0.895	97.8
2022	7.71	3.08	0.868	99.0
2023	7.52	3.16	0.824	95.2
2024	7.11	3.01	0.819	95.6

Each year in this table uses the same 21 stations, with equal station weights. Duration-normalized rainfall is neither an official annual total nor an actual daily total. Eligible monthly and hourly samples can vary.

1. Data sources, fields and cleaning

Sources and fields. Records contain observation and update timestamps, station ID/name/device ID, coordinates, reading update timestamp, rainfall amount, type, unit and date fields. There are no temperature, humidity or wind observations. API previews identify fields and annual IDs; weather findings come from the complete annual files.

Data checks. After downloading complete annual CSV files, we reconcile Spark input row counts with annual API metadata and check duplicate observations, timestamp alignment and rainfall values.

Cleaning. Spark parses timestamps in Asia/Singapore. It accepts finite, nonnegative mm readings with the expected five-minute type and valid station ID, timestamp and CSV record. Timestamps are aligned only if within two seconds of the five-minute grid; original epochs are retained. Station-time duplicates use the latest reading update, then deterministic tie-breakers. Combined annual data are checked again for cross-file duplicates.

Year	Source rows	Duplicates removed	Off-grid over 2 s	Negative readings	Stations
2017	5,256,106	86	0	0	62
2018	4,594,246	0	0	0	56
2019	5,026,862	0	0	0	54
2020	6,334,473	0	0	0	80
2021	7,044,646	0	0	0	76
2022	7,133,230	0	0	0	72
2023	6,742,890	0	0	0	74
2024	6,406,260	0	0	0	70

Cleaning results. All 86 duplicate records occur in 2017. The eight years contain no negative rainfall, timestamps more than two seconds from the five-minute grid, or cross-file duplicates. Missing intervals are not counted as zero rainfall. Additional screening of readings above 50 mm/5min identifies none.

2. Metrics and comparable samples

For a station and time period, let N be the number of valid observations, $r_i \geq 0$ the rainfall amount, $W = \sum_i \mathbf{1}(r_i > 0)$ the number of wet intervals, and E the expected calendar slots. We use:

$$\begin{aligned} \text{Coverage} &= N/E, \qquad E = 288 \times \text{calendar days}, \\ \text{Wet frequency (\%)} &= 100W/N, \\ \text{Wet-interval intensity (mm/5min)} &= \sum_i r_i/W \quad (W > 0), \\ \text{Rain / observed 24h (mm)} &= 288 \sum_i r_i/N. \end{aligned}$$

Calendar denominators handle leap years. Low coverage can reflect a station operating for only part of a year. Intensity averages wet intervals only. For one station and period, normalized rainfall equals 288 × frequency/100 × wet-interval intensity. This product generally does not hold after separate station averaging. Normalized rainfall is not an official daily or annual total and does not reconstruct missed storms.

Fixed annual panel. Of 91 station IDs across eight years, 21 satisfy all criteria: observations in every year, annual coverage of at least 80% in each year, and latitude/longitude spans no larger than 0.005°. The other 70 are excluded only from fixed-panel annual comparisons; their valid observations were not deleted during cleaning. The 21 stations receive equal weights. The next page lists the station scope for each analysis.

Monthly and diurnal samples. Figures 2 and 4 further screen sufficiently complete months or hourly strata from the 21-station pool. Monthly summaries average stations equally, then available years equally. Each eligible year-month has 17–21 stations, and each month has 6–8 available years. Requiring the same station-month to qualify in all eight years leaves only May, June, July, September, October and December, used as an additional comparison.

Spatial samples. Figure 1 uses 2024, which has 70 stations. Of these, 56 have annual coverage of at least 90% and enter map colouring; the other 14 are shown in grey. Figure 5 uses daily totals from the fixed 21 stations, requiring at least 95% daily coverage and 365 common valid days per pair. The 80%, 90% and 95% completeness thresholds are project choices.

3. Computing environment and station scope

Computing environment. Task 1 runs on an Apple Silicon Mac with 10 logical CPUs, 16 GiB RAM, macOS 26.5.2 ARM64, Python 3.12.14, Java Temurin 17.0.20.1+1 and Spark 3.5.8. Task 2 runs on gpushare with Python 3.11.10, Azul Zulu Java 17.0.20.1, Spark 3.5.8 and XGBoost 3.2.0. The gpushare container quota is 42 CPUs and about 667.68 GiB RAM; this project uses two CPUs and a 2 GiB driver heap.

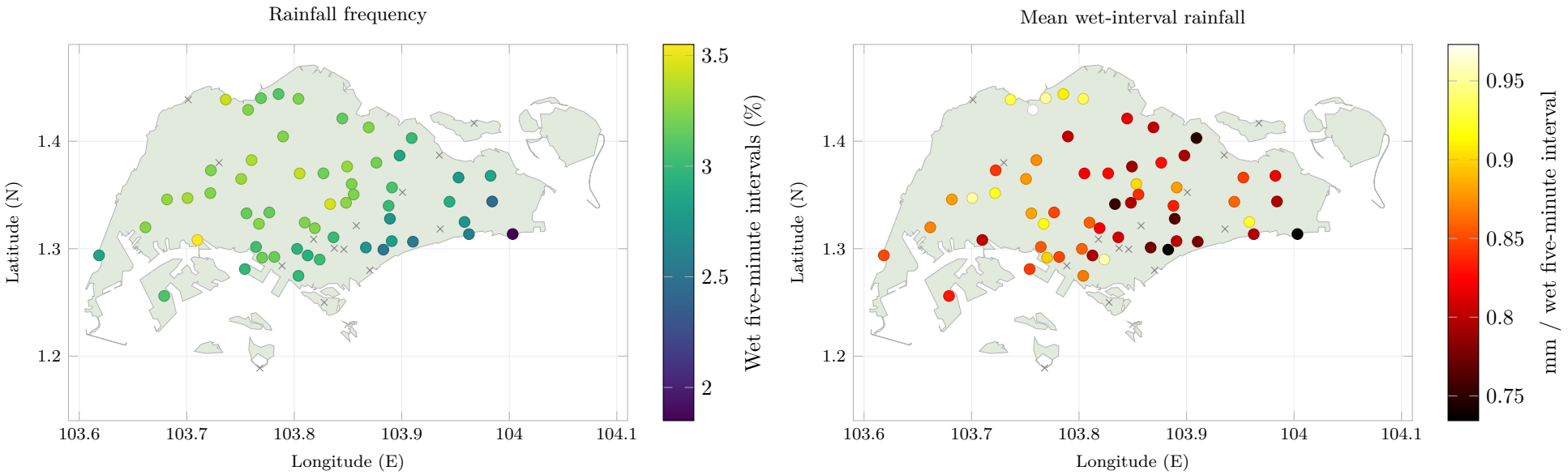
Spark configuration and role. We use single-machine two-thread `local[2]`, with 8 shuffle partitions, default parallelism 2 and no GPU. Spark DataFrame/SQL and windows perform full-data cleaning, temporal aggregation, labels and features; Spark ML pipelines perform model training and evaluation. The gpushare process is bound to CPUs 0 and 1, with sampled benchmark peak memory of 2.719 GiB.

Analysis samples. Station scope and completeness criteria are chosen according to the comparison, as summarized below.

Analysis	Station scope	Screening rule
Figure 1: 2024 station map	56 of 70 stations that year	Annual coverage $\geq 90\%$; 14 shown in grey
Figure 2: monthly patterns	Eligible subsets of the fixed 21	17–21 stations per eligible year-month
Figure 3: annual changes	The same 21 stations	Other 70 excluded from this comparison only
Figure 4: diurnal patterns	Eligible strata of the fixed 21	Year-month-hour coverage $\geq 80\%$
Figure 5: distance and correlation	Fixed 21 stations; 210 pairs	Daily coverage $\geq 95\%$; ≥ 365 shared days
Figure 6 left: wet amounts	All 91 stations across eight years	Wet five-minute observations only
Figure 6 middle/right: exceedance and persistence	Valid 2024 observations from all 70 stations	Six complete future slots or two adjacent slots

Cleaning checks. Source row counts match API counts; 86 duplicate records were removed, and missing slots are not zero-filled. The overall dry fraction uses all 91 stations, rather than only the 21-station panel.

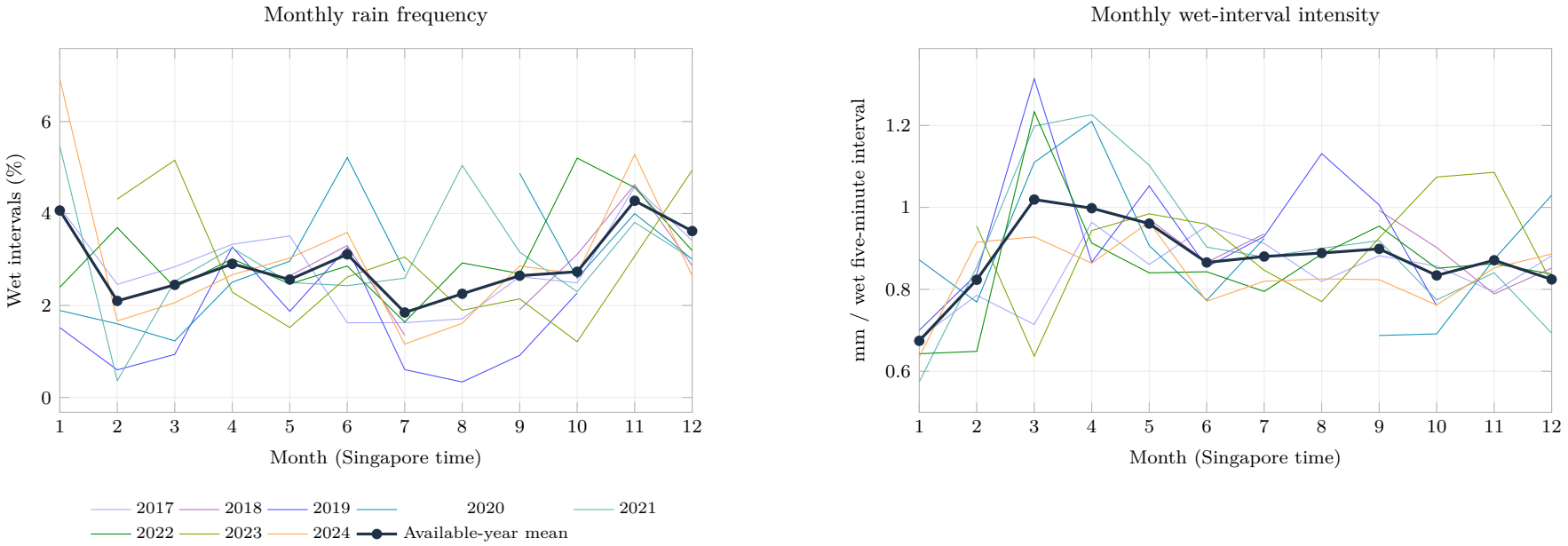
Figure 1: Station rainfall maps in 2024



In 2024, 56 of 70 stations have annual coverage of at least 90%. Coloured points show individual gauge statistics; grey crosses indicate lower coverage. S33 (Jurong Pier Road) has the highest rain frequency, 3.55%. No spatial interpolation is performed. The geoBoundaries land outline is derived from URA 2016 data under ODbL 1.0 and serves as a geographic reference.

Task 2 feature implications (stations). Candidate features include station ID and coordinates to represent station differences. Task 2 should screen windows for historical-feature and future-label completeness, rather than automatically restrict training to the 21-station annual panel.

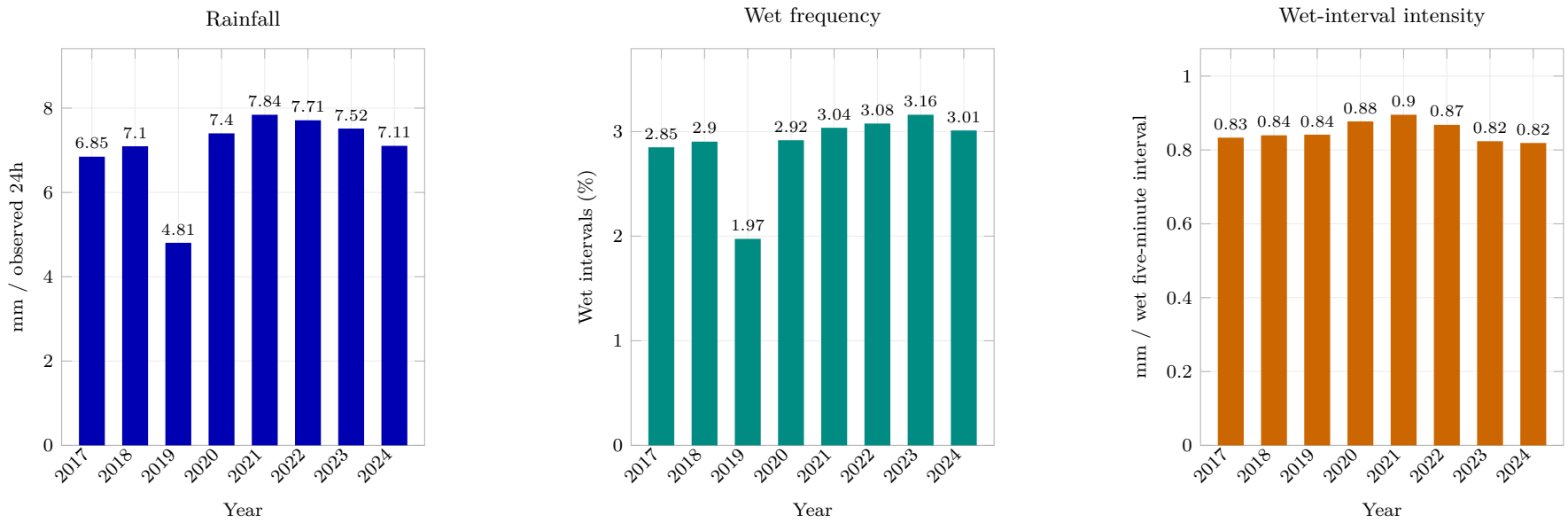
Figure 2: Monthly patterns within a year



Eligible station-months from the fixed panel require at least 80% coverage, with equal averaging over stations followed by available years. Mean monthly rain frequency is highest in November (4.28%) and lowest in July (1.85%). Mean wet-interval amounts are about 1.02 mm in March and 0.67 mm in January. Frequency and wet-interval amount show different seasonal patterns. Each month includes 6–8 years, and each eligible year-month includes 17–21 stations.

Task 2 features (month). Monthly frequency and wet-interval intensity vary differently. We encode month using sine and cosine to represent the annual cycle and combine them with hour encoding to capture seasonal variation in diurnal rainfall.

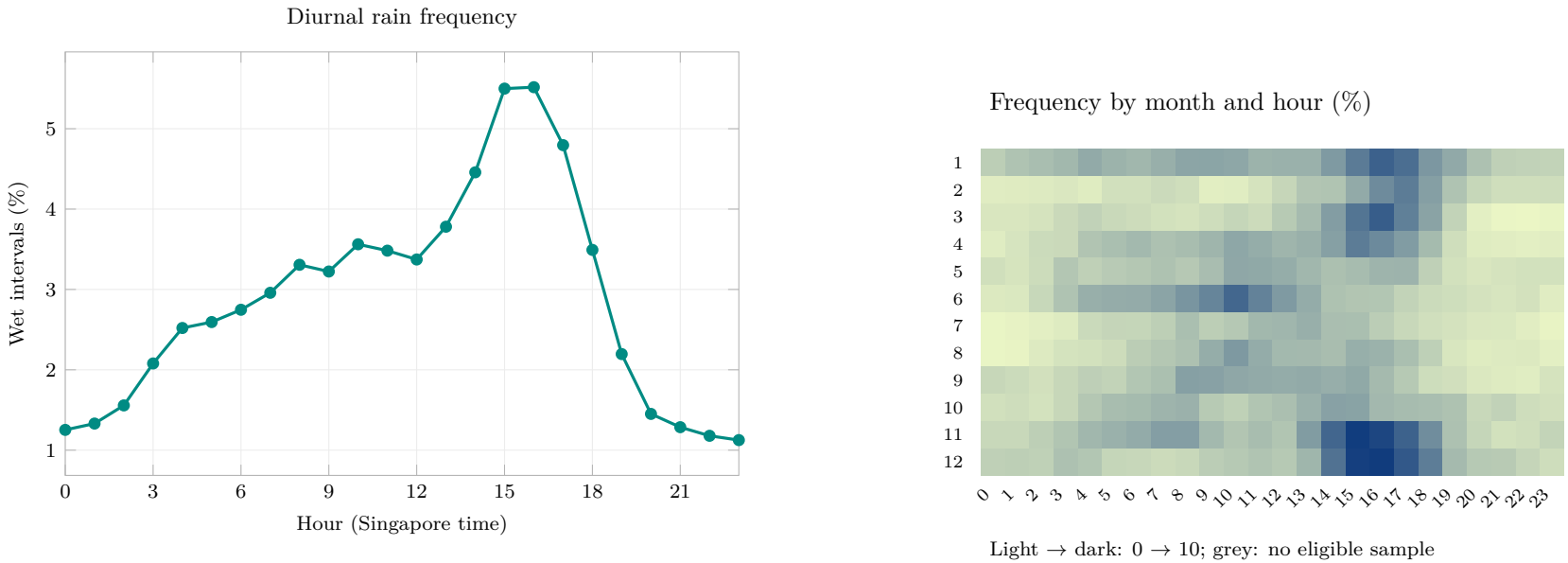
Figure 3: Annual changes in the fixed panel



Each year compares the same 21 equally weighted stations, each with annual coverage of at least 80%. Mean rainfall per observed 24 hours ranges from 4.81 mm in 2019 to 7.84 mm in 2021. Rain frequency in 2019 is 1.97%, below the other years, while its wet-interval amount of 0.841 mm is close to adjacent years. Lower rainfall frequency is thus the main component associated with its lower normalized rainfall.

Task 2 features (year). Training, validation and test are split chronologically. Models use cyclic month and hour features without year identifiers. NEA relates the dry conditions in 2019 to a strong positive Indian Ocean Dipole, providing external context for annual variation.

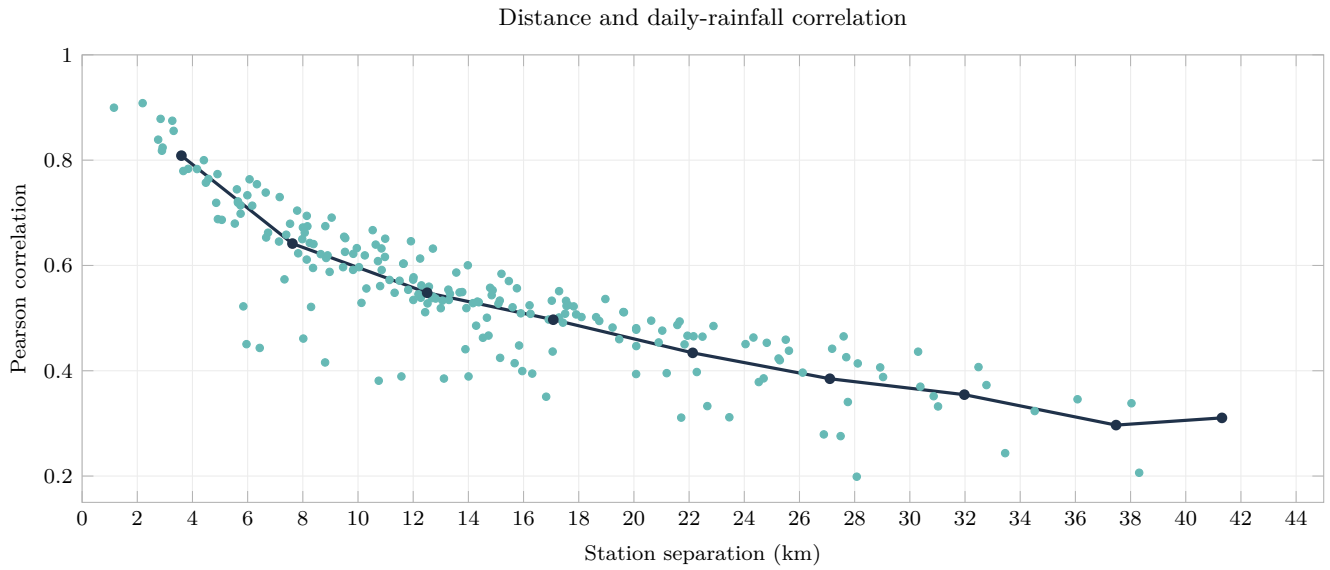
Figure 4: Diurnal variation in rainfall frequency



Rain frequency peaks at 16:00–17:00, with peak timing and strength varying by month. The afternoon peak is more pronounced in November–December, while June has elevated frequency from morning toward midday. Statistics equally weight eligible station-year-month-hour strata with at least 80% coverage. These findings motivate hour, month and their interactions as predictors.

Task 2 feature implications (hour and month). Encode Singapore local hour cyclically and consider month–hour interactions: the strength and timing of the afternoon peak vary by month. Logistic regression and linear SVM need explicit interaction features; trees can learn such combinations. Their predictive contribution must still be tested by ablation.

Figure 5: Distance and daily-rainfall correlation

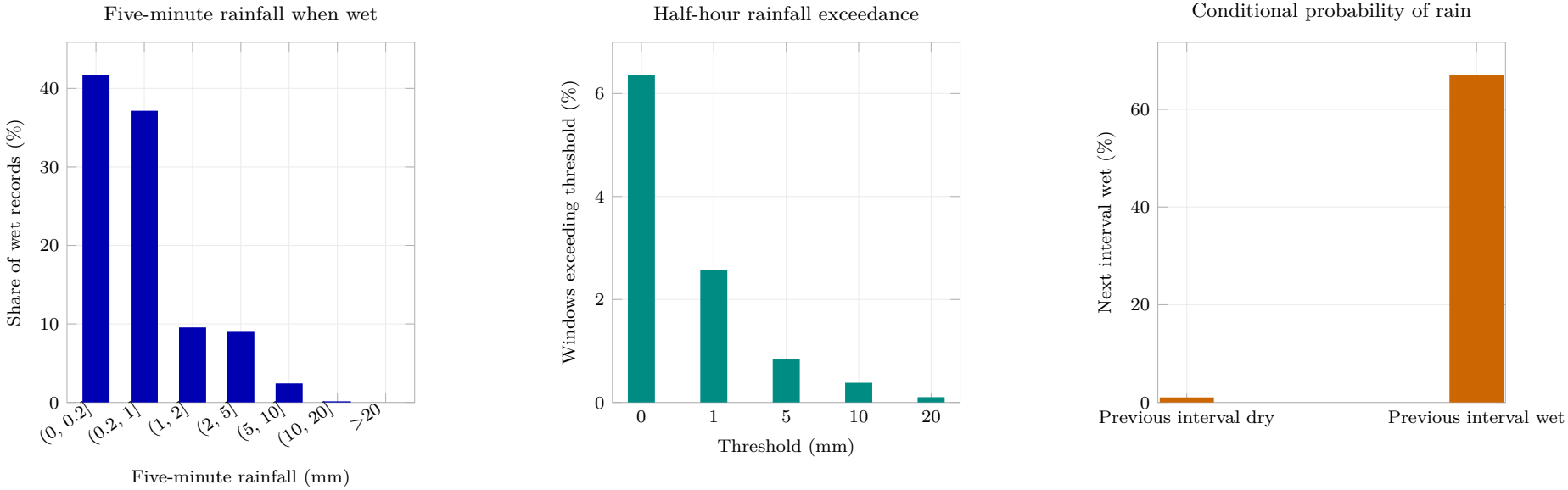


The 21 stations form 210 pairs. The mean Pearson correlation of daily rainfall is 0.809 for 17 pairs separated by at most 5 km, compared with 0.336 for 12 pairs more than 30 km apart. Mean correlations in 5 km distance groups generally decrease with separation, indicating spatial dependence.

Stations are not independent. Nearby gauges may experience the same rainfall event, and different pairs can share a station. Thus, 91 gauges are not 91 independent weather samples, and the 210 correlations are not 210 independent samples. Shared-day screening follows the criteria given earlier.

Task 2 features (neighbours). Stronger daily-rainfall correlation among nearby stations motivates mean rainfall, maximum rainfall and wet fractions from the three closest stations. Task 2 compares models before and after adding neighbour features to evaluate their contribution to half-hour prediction.

Figure 6: Rainfall distribution, half-hour exceedance and persistence



Rainfall distribution in wet intervals. Across all stations and eight years, 41.72% of wet five-minute observations fall in $(0, 0.2]$ mm and 37.16% in $(0.2, 1]$ mm. Together, 78.88% have at most 1 mm, indicating small accumulations in most wet intervals. Dry intervals account for 97.08% of all valid five-minute observations.

Exceedance probability of half-hour rainfall. There are 6,374,476 complete future six-interval windows in 2024. Subsequent 30-minute totals strictly exceed 0 mm in 6.36% of windows and 5 mm in 0.83%; exceedance becomes less frequent as the rainfall threshold rises. Totals are compared at nine decimal places to avoid floating-point errors at threshold boundaries. The horizontal axis uses threshold categories.

Rainfall persistence in consecutive intervals. For consecutive five-minute observations at the same station in 2024, the probability of rain is 1.04% following a dry interval and 67.03% following a wet interval. This substantial short-term persistence supports recent rainfall and wet-spell duration as predictors.

Task 2 feature implications (persistence and rare events). Implemented features include the latest reading, 15/30/60-minute totals, recent rainfall changes and wet-spell duration. Only records available at prediction time are used; missing readings are not dry readings. The following sections report positive counts, PR-AUC, Brier score and calibration curves separately by threshold.

4. Task 2: prediction target and chronological splits

Prediction target. For station s, prediction time t and rainfall threshold q, the model estimates the probability that the following 30-minute accumulated rainfall strictly exceeds q. Historical station and neighbour features are retrieved for the requested station and time.

$$R^{30}_{s,t} = \sum_{k=1}^{6} r_{s,t+5k\,\min}, \qquad y^{(q)}_{s,t} = \mathbf{1}(R^{30}_{s,t} > q), \qquad \hat{p}_q = P(y^{(q)} = 1 \mid x_{s,t}).$$

Rainfall thresholds. We use $q \in \{0, 5, 25\}$ mm for three targets: any rain, accumulation above 5 mm and accumulation above 25 mm. One classifier is trained for each target. Each station-time row retains one feature set and three label columns, and prediction uses the model corresponding to the input threshold.

Label construction. Taking timestamps as five-minute interval ends, sum the six valid rainfall readings in $(t, t + 30\text{min}]$ and compare the total at nine decimal places with the threshold. The current interval is excluded from future accumulation. A label is constructed only when all six future intervals have valid readings. Equality with the threshold is labelled zero.

Purpose	Valid rows	> 0 positives	> 5 positives	> 25 positives
2017–2022: eligible training pool	34,542,285	–	–	–
2017–2022: training sample (about 5%)	1,727,966	103,477	13,712	926
2023 H1: probability calibration	3,325,499	251,794	30,699	1,532
2023 H2: model selection	3,393,373	197,358	28,000	2,487
2024: final test	6,374,476	405,430	53,193	3,517

Training sample. We select approximately 5% of the 34,542,285 eligible training rows from 2017–2022, yielding 1,727,966 observations. Inclusion is determined by a hash of station and time with seed 5208, and all comparison models use the same sample. The first half of 2023 fits probability-calibration functions; the second half compares models. The 2024 data form the test set.

5. From Task 1 findings to feature engineering

Task 1 motivation	Construction	Implemented inputs
Afternoon peak varies by month	Cyclic local hour/month and four interactions	8 calendar features; no year ID
Station differences and spatial correlation	Station ID, coordinates, three closest neighbours	Neighbour means, maxima, wet fractions
Short-term rainfall persistence	Latest reading and 5/10/15/30/60-min lags	15/30/60-min totals, changes, wet-spell length

Historical features. Use observations in the preceding 90 minutes with observation and stored-update times no later than prediction time. A latest reading older than 30 minutes is marked unavailable. Lagged rainfall is retrieved on the fixed five-minute grid. Totals over 15/30/60 minutes end at the latest available reading, accompanied by observation counts and reading age.

Neighbour construction. Choose the three closest other stations using coordinates known by prediction time. Disable a station spatial contribution if it moves more than 0.005° from its initial location. Join only readings available as of t. Aggregate latest amounts and past 30-minute totals into means, maxima and available counts, plus wet fractions, mean distances and ages. Spark handles full-data time windows, features and neighbour joins; a small coordinate table supports distance calculations.

Time encoding. Let h denote local hour including fractional minutes and m denote month. Hour features are $\sin(2\pi h/24)$ and $\cos(2\pi h/24)$; month features are $\sin[2\pi(m-1)/12]$ and $\cos[2\pi(m-1)/12]$. Their four pairwise products give interactions, for eight time features in total. Cyclic encoding keeps neighbouring 23:00 and 00:00, and December and January, close in feature space.

Model-input processing. Linear models one-hot encode station ID; tree models treat station ID as categorical. Missing numeric values are filled with training-sample means and accompanied by missing indicators. Constant values are excluded. Final input dimensions are 162 for linear models and 73 for trees. Logistic regression and SVM use feature standardization during fitting.

L2 logistic feature ablation	Test PR-AUC: 0 mm	5 mm	25 mm
Calendar and station	0.11652	0.01647	0.00136
Add own-station history	0.53061	0.15788	0.01485
Add neighbours	0.55978	0.17909	0.01734

Feature-group comparison. L2 logistic models compare three progressively expanded inputs: time and station, added own-station history, and added neighbour history. Each group trains three targets, for nine models. This initial comparison shares the training sample and full 2023 validation, uses an 80-iteration cap and no probability calibration. Own-station history substantially improves test PR-AUC for all targets, and neighbour history provides further gains. Subsequent model comparisons therefore use the complete feature set.

6. Models, imbalance policies and probability calibration

Model	Fixed parameters
L2 logistic (MLlib)	$\lambda = 0.001$, standardization, 100 iterations, tolerance 10^{-6}
Linear SVM (MLlib)	$\lambda = 0.001$, standardization, 100 iterations, tolerance 10^{-6}
GBT (MLlib)	40 rounds, depth 4, step 0.1, 128 bins, minimum 20 rows/node
XGBoost (Spark interface)	80 trees, depth 4, learning rate 0.1, 64 bins, child weight 20; $L_2 = 1$

Implementation. Logistic, linear SVM and GBT use `pyspark.ml`; XGBoost uses `SparkXGBClassifier` with two CPU workers. All models use the fixed parameters listed above and random seed 5208. Nine SVM and two weighted logistic models reach the 100-iteration cap.

Three training policies. Natural distribution assigns unit weights. Balanced weighting uses $w_+ = N/(2P)$ and $w_- = N/[2(N-P)]$, giving equal class mass and overall mean weight one. Constrained SMOTE targets 20% positives, caps their total at ten times the original positives, and augments training only. Calibration, selection and test are neither resampled nor class-weighted.

Constrained SMOTE. To retain relationships among station, time and historical rainfall, search up to five neighbours among minority examples with the same station, missingness, discrete states and rainfall-zero pattern. Interpolate 16 continuous rainfall features using one coefficient; time encoding, coordinates, observation counts, wet fractions and ages stay fixed. Examples without suitable neighbours are duplicated. Generated features satisfy nonnegative rainfall, ordering of complete totals and wet-count consistency. Single-thread sklearn searches training minority examples; other data processing and training use Spark.

Threshold mm	Added rows	Augmented positives (%)	Rows with changed rain features	Isolated-profile copies
0	302,646	20.000	129,231	73,685
5	123,408	7.406	34,344	48,033
25	8,334	0.533	773	6,061

Probability calibration (Platt method). After training each base classifier, fit a score-to-probability function on 2023 H1 observations: $\hat{p} = 1/[1 + \exp(-(az + b))]$, where z is the raw model score and a, b are fitted parameters with $a \geq 0$. This converts SVM scores into probabilities and adjusts probability scales after weighting or SMOTE. The aim is agreement between predicted probabilities and observed frequencies: for example, approximately 10% of examples assigned about 10% probability should be positive. The calibration function is fitted using MLlib logistic regression with regularization 10^{-6}.

7. Evaluation protocol and validation selection

Evaluation metrics. ROC-AUC measures positive-negative ranking, with random-ranking reference 0.5. PR-AUC combines precision and recall and is useful when positives are rare. Under random ranking, precision at different recall levels approaches positive prevalence π, giving a PR-area reference near π. Test prevalences are 0.06360, 0.008345 and 0.0005517. Both ROC-AUC and PR-AUC use Spark integration over exact score ranks.

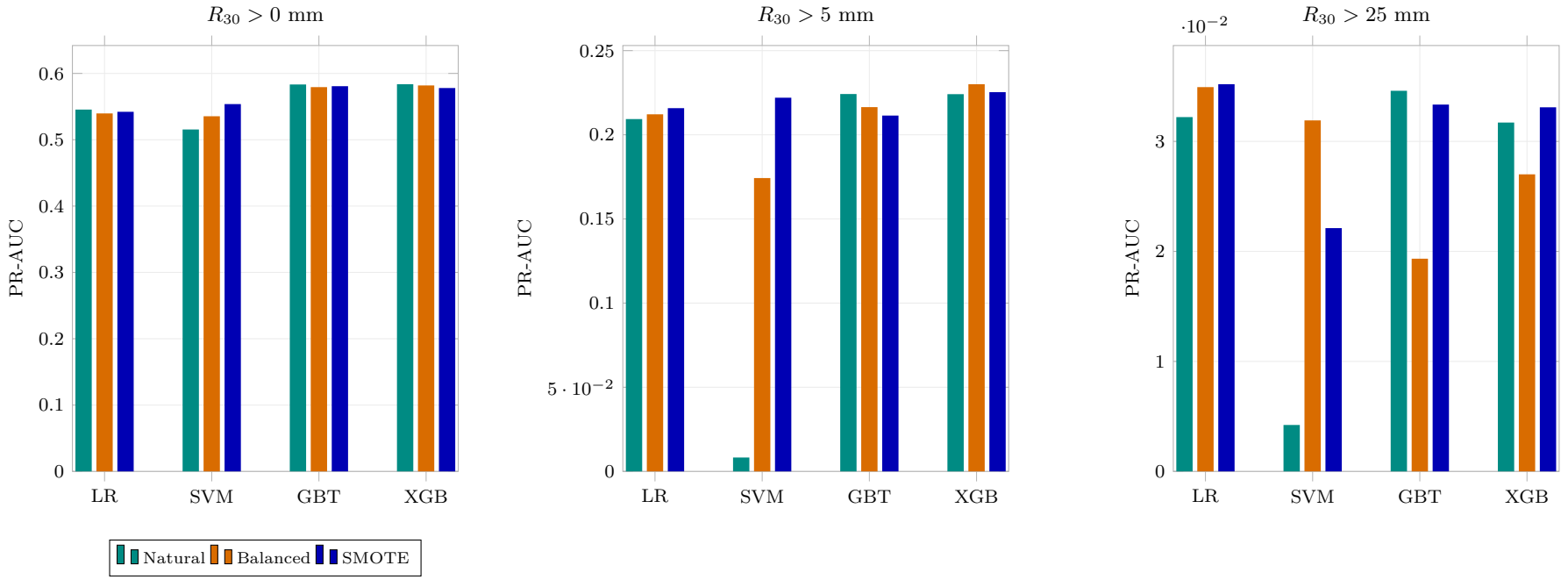
$$\text{Brier} = \frac{1}{n}\sum_{i=1}^{n}(\hat{p}_i - y_i)^2, \qquad \text{Precision} = \frac{TP}{TP+FP}, \quad \text{Recall} = \frac{TP}{TP+FN}.$$

Probability evaluation. Brier score is mean squared error between predicted probabilities and labels; smaller is better. The baseline always predicts training-sample prevalence. Reliability plots group test examples with similar predictions and compare mean predicted probability on the horizontal axis with actual positive frequency on the vertical axis; ideal agreement is $y = x$. A validation-selected probability cutoff maximizing F1 supplies precision, recall, F1 and accuracy. This probability cutoff acts on model output and differs from the rainfall threshold q defining labels.

Algorithm / policy	Selection PR-AUC: 0 mm	5 mm	25 mm
Logistic / Natural	0.54547	0.20932	0.03221
Logistic / Balanced	0.53968	0.21216	0.03493
Logistic / SMOTE	0.54213	0.21581	0.03520
SVM / Natural	0.51546	0.00825	0.00422
SVM / Balanced	0.53541	0.17428	0.03191
SVM / SMOTE	0.55375	0.22202	0.02212
GBT / Natural	0.58332	0.22421	0.03460
GBT / Balanced	0.57932	0.21642	0.01934
GBT / SMOTE	0.58073	0.21141	0.03335
XGBoost / Natural	0.58368	0.22413	0.03171
XGBoost / Balanced	0.58186	0.23002	0.02701
XGBoost / SMOTE	0.57796	0.22531	0.03310

Model selection. Compare training policies for each algorithm and rainfall threshold by 2023 H2 validation PR-AUC, using Brier to resolve ties. The final models are naturally trained XGBoost for 0 mm, class-weighted XGBoost for 5 mm, and SMOTE logistic for 25 mm. All 12 models selected across algorithms and targets are evaluated on full 2024 test data.

Figure 7: imbalance policies on the selection set



Validation findings. Trees outperform linear models in PR-AUC at 0 mm. Balanced XGBoost performs best at 5 mm, while SMOTE logistic performs best at 25 mm. SMOTE improves some linear models, whereas GBT obtains its highest validation PR-AUC with natural training for all targets. Imbalance-policy benefits vary by algorithm and target.

8. Test results in 2024

Rainfall threshold mm	Algorithm	Selected policy	ROC-AUC	PR-AUC	Brier
0	Logistic	Natural	0.85500	0.55978	0.0392757
0	SVM	SMOTE	0.86049	0.57387	0.0399706
0	GBT	Natural	0.87749	0.60338	0.0368843
0	XGBoost	Natural	0.87986	0.60797	0.0364959
5	Logistic	SMOTE	0.85004	0.18162	0.0075438
5	SVM	SMOTE	0.77601	0.20093	0.0077220
5	GBT	Natural	0.86100	0.20891	0.0074329
5	XGBoost	Balanced	0.87526	0.21299	0.0072292
25	Logistic	SMOTE	0.85651	0.01805	0.0005483
25	SVM	Balanced	0.84840	0.01550	0.0005484
25	GBT	Natural	0.76516	0.01881	0.0005618
25	XGBoost	SMOTE	0.88889	0.01918	0.0005461

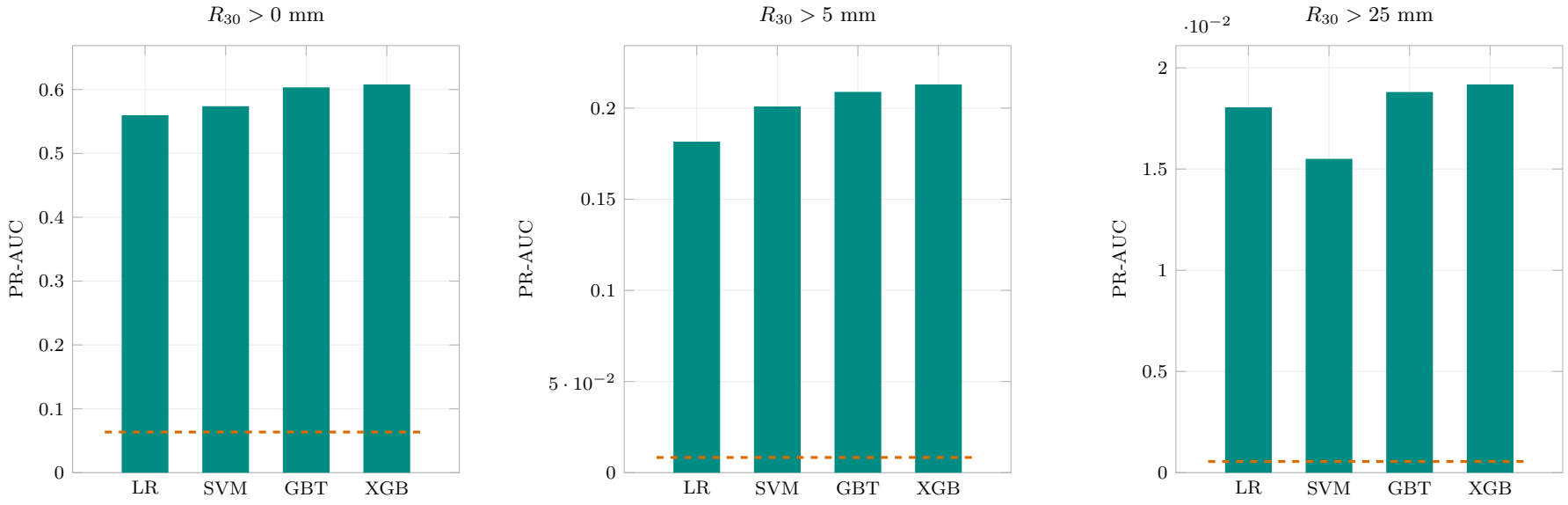
Test size and baselines. Each target is evaluated on 6,374,476 real windows with prevalences of 6.3602%, 0.83447% and 0.055173%. Constant training-prevalence Brier baselines are 0.0595707, 0.0082752 and 0.0005514. Always predicting negative gives accuracies of 93.64%, 99.17% and 99.945%, showing why accuracy cannot replace rare-event detection. All table results use calibrated probabilities.

Test findings. XGBoost test PR-AUC is 0.60797, 0.21299 and 0.01918 across the three targets. Its 25 mm ROC-AUC is 0.88889 while PR-AUC remains low, reflecting the difficulty of detecting rare high-rainfall events. At this threshold, GBT Brier is 0.0005618, above the constant baseline, showing that ranking and probability accuracy need separate assessment. The final 25 mm model remains validation-selected SMOTE logistic.

Rainfall threshold mm	Algorithm	Training PR-AUC	Validation PR-AUC	Test PR-AUC
0	XGB	0.61237	0.58368	0.60797
5	XGB	0.22789	0.23002	0.21299
25	LOGISTIC	0.04515	0.03520	0.01805

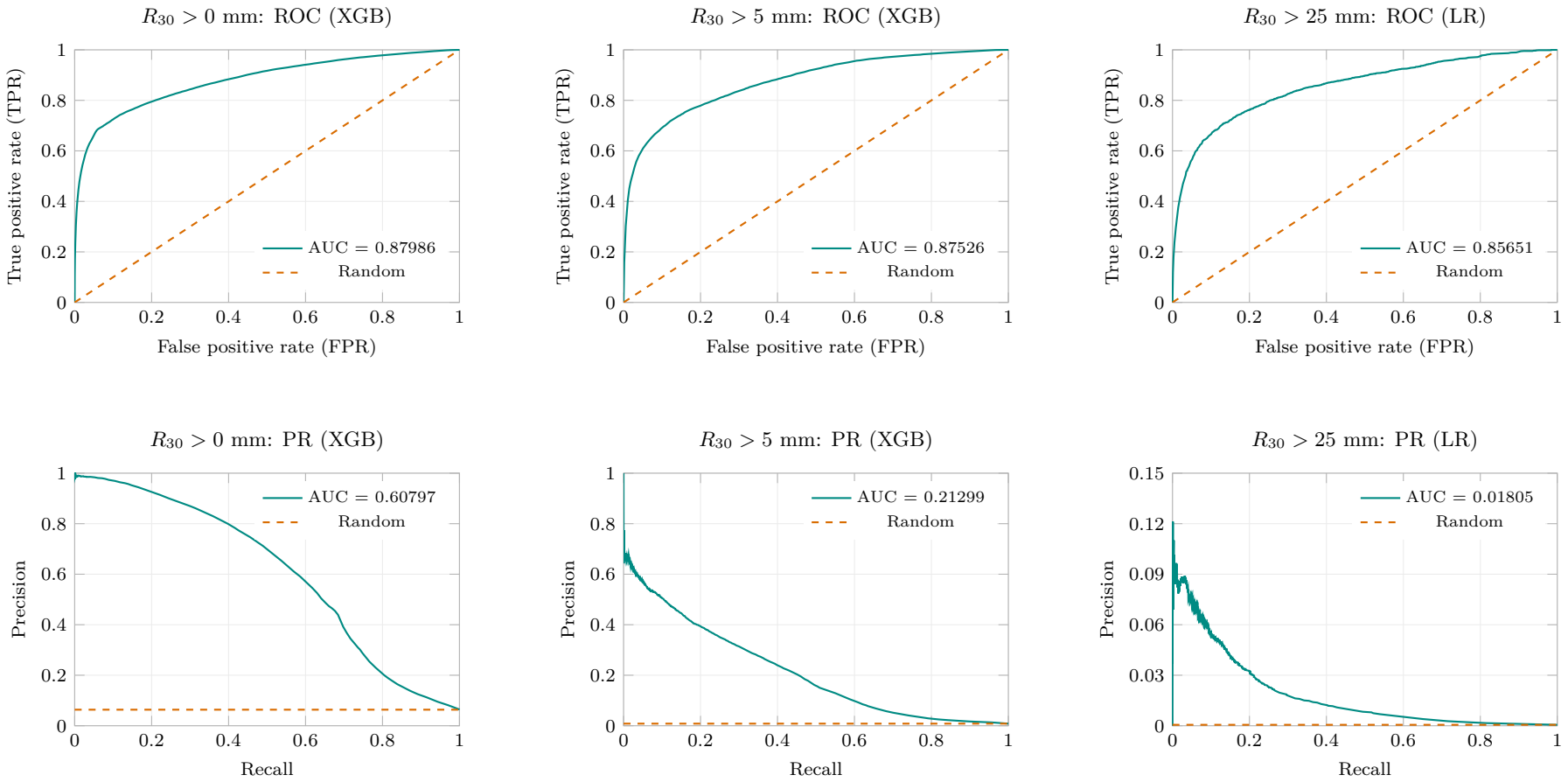
Training and test comparison. Training metrics use a fixed 20% subset of original training observations (344,957 rows), while validation and test use all eligible rows. Naturally trained GBT at 25 mm has training PR-AUC 0.17452 and validation PR-AUC 0.03460, a substantial gap. Positive prevalence differs by period, so changes in PR-AUC also reflect event frequency.

Figure 8: test ranking and probability error



Model comparison. XGBoost obtains the highest test PR-AUC for all three targets. The orange dashed line denotes the approximate random-ranking reference, equal to test prevalence. At 25 mm, PR-AUC exceeds the roughly 0.00055 reference but remains low in absolute terms, making rare-event prediction the most difficult target.

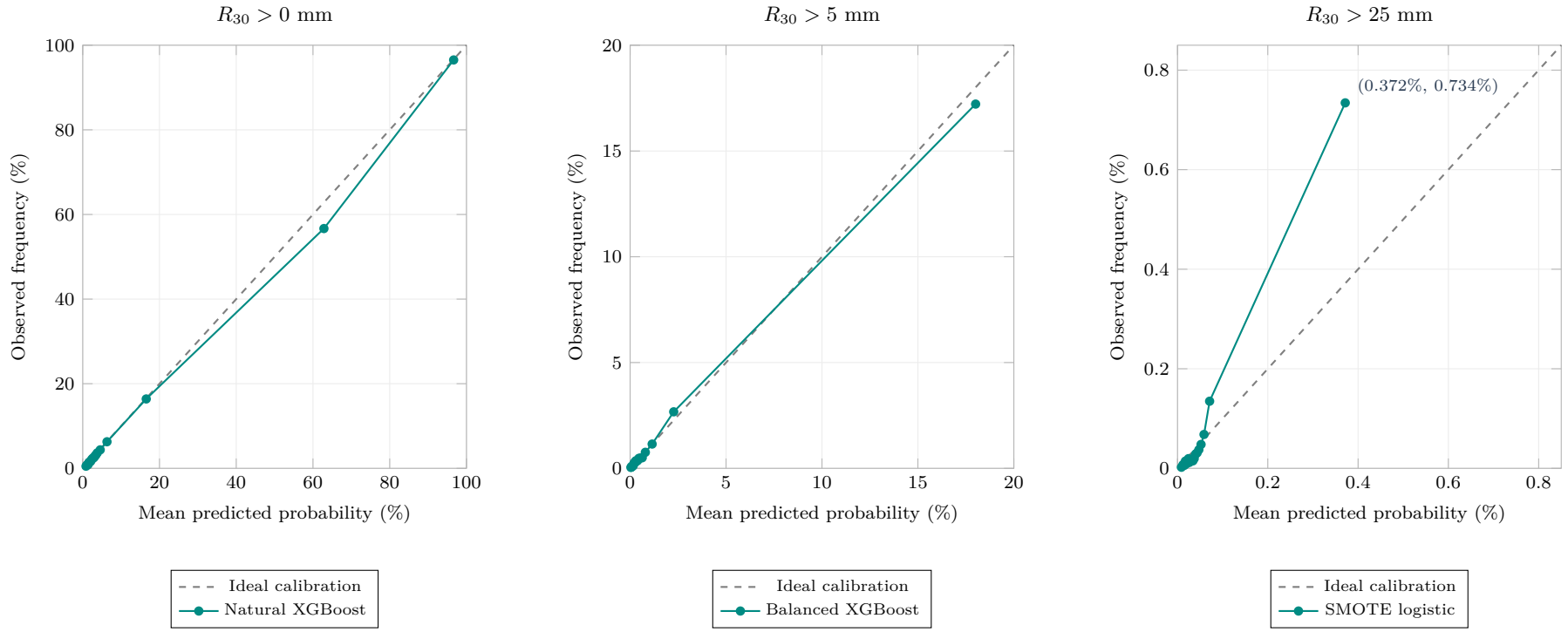
Figure 9: ROC and PR curves of the final selected models



Evaluation scope. The columns show natural-distribution XGBoost, class-weighted XGBoost and SMOTE logistic, respectively, selected using 2023 H2 validation. Curves and AUCs use all 6,374,476 eligible 2024 test windows; models use the previously described training sample of about 5%. Dashed references are $y = x$ for ROC and test prevalence for PR. The 25 mm PR panel magnifies the precision axis to 0–0.15.

Ranking performance. ROC-AUCs are 0.87986, 0.87526 and 0.85651; PR-AUCs are 0.60797, 0.21299 and 0.01805. PR curves decline as the rainfall threshold increases, indicating greater difficulty in achieving high precision and recall together for rarer events. Despite a relatively high ROC-AUC, detection of rare 25 mm events remains limited.

Figure 10: reliability curves and probability quality



Test calibration findings. The plots show the final selected model for each rainfall threshold. At 0 mm, the XGBoost group with mean predicted probability 96.66% has observed frequency 96.52%. At 5 mm, the balanced XGBoost group with mean prediction 18.01% has observed frequency 17.22%. For SMOTE logistic at 25 mm, the highest-prediction group contains 252,650 windows, with mean prediction 0.372% versus actual exceedance frequency 0.734%, indicating underestimated risk in this group. All axes use percentages.

Before and after calibration. For SMOTE logistic at 25 mm, validation Brier falls from 0.0011352 before calibration to 0.0007248 afterward. Mean predicted probability falls from 0.55317% to 0.04886%, against actual prevalence of 0.07329%. Platt calibration reduces overall probability inflation after resampling, while higher-risk groups remain underpredicted.

9. Classification metrics and further analysis

Rainfall threshold mm	Validation-selected probability cutoff	Precision	Recall	F1	Accuracy
0	0.31989	0.61636	0.56524	0.58970	0.94997
5	0.15488	0.28201	0.34305	0.30955	0.98723
25	0.01462	0.05165	0.11288	0.07087	0.99837

Classification findings. Test classification metrics use probability cutoffs chosen on validation. At 25 mm, low F1 coexists with high accuracy because negatives dominate. We therefore compare models primarily by ROC-AUC and PR-AUC, and evaluate probability predictions using Brier score and reliability curves.

SMOTE findings. At 25 mm, training gains 8,334 examples; 773 change rainfall features and 6,061 are duplicates without suitable neighbours. Strict constraints leave considerable repeated information, which may explain limited gains. More real high-rainfall examples and tuned class weights or model parameters are possible improvements.

Probability ordering across thresholds. Separately trained targets can assign higher probability to a higher rainfall threshold. Among algorithm-specific selected models on 2024 data, violations of $\hat{p}_{25} \leq \hat{p}_5 \leq \hat{p}_0$ occur in 1 logistic, 2,448 SVM, 528 GBT and 247 XGBoost windows, each out of 6,374,476. These counts compare three models from the same algorithm. Joint constraints on the final mixed-algorithm combination could be addressed with monotone calibration.

Further analysis. The same rainfall event can affect neighbouring stations, and consecutive half-hour windows overlap. Future evaluation could group by date or rainfall event to examine stability across stations, years and events.

10. AI use and references

AI-use declaration. We use OpenAI Codex to assist study design, data-processing and model code, result checks and report editing. Reported rainfall statistics and model evaluations come from actual processing and model runs; SMOTE examples are used only for training.

Sources.

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